# National University of Computer and Emerging Sciences, Lahore Campus

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| **Course:** | **Data Warehousing** | **Course Code:** | |
| **Program:** | **BS** | **Semester:** | |
| **Practice Problem3:** | **Indexing Techniques - Solution** | | |

**Instruction/Notes:**

Consider the following tables and statistics which are part of a Library system:
***Book*** *(BookID, Title, Publisher, PublishYear, Author, …);*

Assume Book table containing one *million* rows. Table row and index entry take *150 bytes* and 15 *bytes of* space respectively. Data block size is 16*KB* and available memory size is *100 blocks*. Suppose selectivity of publisher 'Pearson'= 5%, publisher 'Wiley'= 3%, publishYear '2018'= 3%, publishYear '2020'= 2%, and publishYear '2022'= 1%.

**Q.** Calculate the total I/O cost for the Query using the following indexing techniques. Show all steps clearly.

**Query:** *SELECT * FROM book WHERE publisher IN ('Pearson', 'Wiley') AND publishYear IN ('2018', '2020', '2022');*

**1.** Single index access *(Assume Single indexes exist on Publisher and PublishYear columns separately)*
**2.** Combining multiple indexes *(Assume Single indexes exist on Publisher and PublishYear columns separately)*
**3.** Composite Index access *(Assume a composite index exist on Publisher and PublishYear columns, with index entry size=15 bytes)*

**Answer:** ***Combine selectivity of account is 8% of (6% off 1,000,000) = 4800 rows.***

**K**=100; **B**=16k (i.e. 16,384); **R**=150; $\mathbf{R_i}$=15; **bfr**=109 (i.e. $\lfloor B/R \rfloor$=16k/150); $\mathbf{bfr_i}$=**1092** (i.e. $\lfloor B/R_i \rfloor$=16k/15);

$r_B$=1m; $\mathbf{b_B}$=**9175** (i.e. $\lceil r_B/bfr \rceil$= 1m/109); $\mathbf{b_{Bi}}$=**916** (i.e. $\lceil r_B/bfr_i \rceil$= 1m/1092);

**1. Single index access:**

Index cost = Choose highest selectivity column (i.e. publishYear - 6% of rows): 6% of $1m/bfr_i$ = 60,000/1092 = **55**
As qualifying rows are greater than total no of blocks of book table (i.e. 60,000 > 9175), so we have to read all the blocks of book table.
Total cost = Index access cost + base table access cost
= 55 + 9175 = **9230** I/Os

**2. Combining multiple indexes:**

Publisher Index cost = 8% of $1m/bfr_i$ = 80,000/1092 = 74
PublishYear Index cost = 6% of $1m/bfr_i$ = 60,000/1092 = 55

Total cost = Indexes access cost + base table access cost
= (74+55) + 4800 = **4929** I/O

**3. Composite index access:**

Combination#1 (Pearson, 2018): 5% of (3% of 1m)= 1500 RIDs and 2 I/Os (i.e. $RIDs/bfr_i$ = 1500/1092)
Combination#2 (Pearson, 2020): 5% of (2% of 1m)= 1000 RIDs and 1 I/O (i.e. $RIDs/bfr_i$ = 1000/1092)
Combination#3 (Pearson, 2022): 5% of (1% of 1m)= 500 RIDs and 1 I/O (i.e. $RIDs/bfr_i$ = 500/1092)
Combination#4 (Wiley, 2018): 3% of (3% of 1m)= 900 RIDs and 1 I/O (i.e. $RIDs/bfr_i$ = 900/1092)
Combination#5 (Wiley, 2020): 3% of (2% of 1m)= 600 RIDs and 1 I/O (i.e. $RIDs/bfr_i$ = 600/1092)
Combination#6 (Wiley, 2022): 3% of (1% of 1m)= 300 RIDs and 1 I/O (i.e. $RIDs/bfr_i$ = 300/1092)
Total Index access cost = 2+1+1+1+1+1 = **7**
Total I/Os (Index access cost + base table access cost) = 7 + 4800 blocks